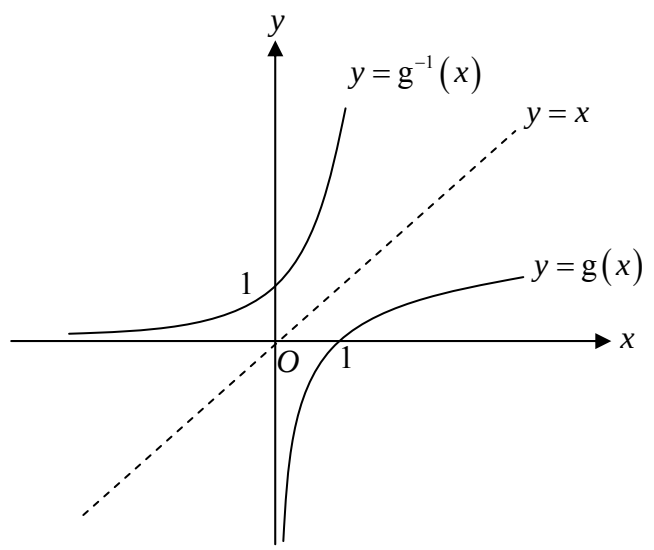
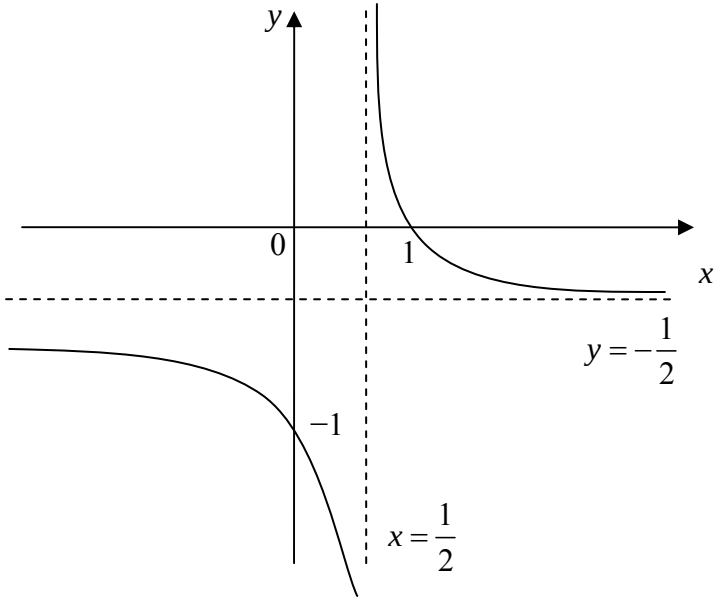
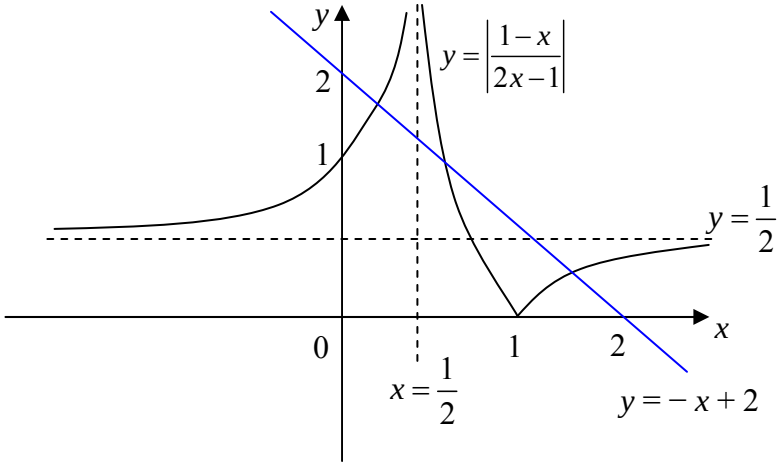
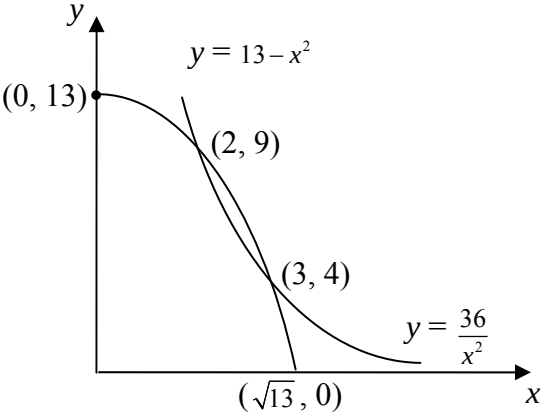


2008 H1 MATHS (8863) JC2 PRELIMINARY EXAM - MARKING SCHEME

Qn	Solution
1	Simultaneous Equation
	Sub. (1,2) into equations: $a + 2b = 2$ --- (1) $b + 2a^2 = 10$ --- (2) From (1), $a = 2 - 2b$ ---(3) Sub. (3) into (2): $b + 2(2 - 2b)^2 = 10$ $8b^2 - 15b - 2 = 0$ $(b - 2)(8b + 1) = 0$ $b = 2$, $a = -2$ or $b = -\frac{1}{8}$, $a = \frac{9}{4}$ (rejected $a > 0$)

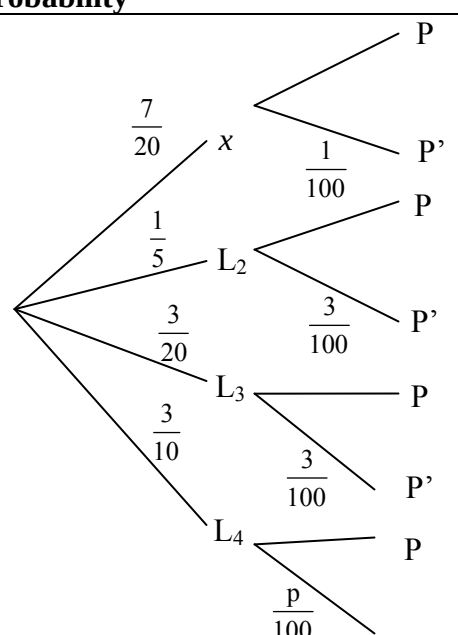
Qn	Solution
2	Functions
(i)	
(ii)	$R_g = (-\infty, \infty), D_f(0, \infty)$ For fg to exist, $R_g \subseteq D_f$, largest domain of g is $D_g = (1, \infty)$. $fg(x) = \cos(\ln(x^2))$. $D_g = (1, \infty) \rightarrow R_g = (0, \infty) = D_f \rightarrow R_{fg} = [-1, 1]$ $\therefore R_{fg} = [-1, 1]$

Qn	Solution
3	Curve Sketching
(i)	$y = \frac{1-x}{2x-1} = \frac{-\frac{1}{2}(2x-1) + \frac{1}{2}}{2x-1} = -\frac{1}{2} + \frac{1}{2(2x-1)}$ Equations of asymptotes are $x = \frac{1}{2}$ and $y = -\frac{1}{2}$. 
(ii)	 $ 1-x = (-x+2) 2x-1 $ $\left \frac{1-x}{2x-1} \right = -x+2$ The line $y = -x + 2$ intersects the curve $y = \left \frac{1-x}{2x-1} \right$ at exactly three points. Hence the equation $1-x = (-x+2) 2x-1$ has 3 real roots.

Qn	Solution
4	Integration and Applications
	(i)  (ii) $\int \frac{36}{x^2} dx = -\frac{36}{x} + C$ $\int (13 - x^2) dx = 13x - \frac{x^3}{3} + C$ (iii) $\int_2^3 \frac{36}{x^2} dx$ $= \left[-\frac{36}{x} \right]_2^3$ $= -\frac{36}{3} - \left(-\frac{36}{2} \right)$ $= 6$ $\int_2^3 (13 - x^2) dx$ $= \left[13x - \frac{x^3}{3} \right]_2^3$ $= 13(3) - \frac{3^3}{3} - \left(13(2) - \frac{2^3}{3} \right)$ $= \frac{20}{3}$ Area = $\frac{20}{3} - 6 = \frac{2}{3}$

Qn	Solution
5	Differentiation and Applications
(a)	$\frac{dy}{dx} = 12x^2 + 6$ (i) when $x = 1$, $\frac{dy}{dx} = 18$ $y - 10 = 18(x - 1)$ $y = 18x - 8$ (ii) $y + 10 = -\frac{1}{18}(x + 1)$ $y = -\frac{1}{18}x - \frac{181}{18}$
(b)	$2x + 2y + \frac{1}{2}(2\pi x) = 20 \quad \text{where } y = \text{length of } RS = \text{length of } PT$ $y = \frac{1}{2}(20 - 2x - \pi x)$ total area, $A = (2x)(y) + \frac{1}{2}(\pi x^2)$ $= (2x)\left(\frac{20 - 2x - \pi x}{2}\right) + \frac{\pi x^2}{2}$ $= 20x - 2x^2 - \frac{1}{2}\pi x^2$ $= 20x - \frac{x^2}{2}(\pi + 4)$ For maximum Area, $\frac{dA}{dx} = 20 - (\pi + 4)x = 0$ $\Rightarrow x = \frac{20}{\pi + 4}$

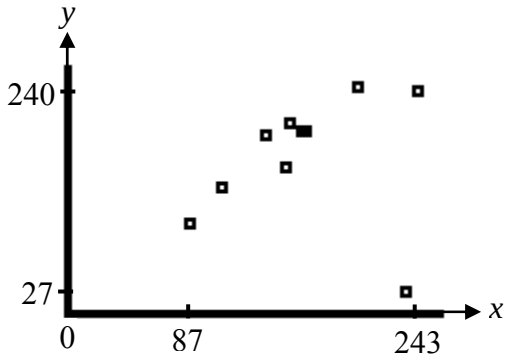
Qn	Solution
6	Sampling Methods
(i)	Systematic sampling
(ii)	Advantage: 1. Evenly spread over each shift 2. Easy to conduct Disadvantage: The different flavours are not evenly distributed.
(iii)	$\text{Beef} = \frac{170}{500} \times 50 \approx 17 \text{ packets}$ $\text{Chicken} = \frac{120}{500} \times 50 \approx 12 \text{ packets}$ $\text{Beef with rice} = \frac{150}{500} \times 50 \approx 15 \text{ packets}$ $\text{Chicken with rice} = \frac{60}{500} \times 50 \approx 6 \text{ packets}$ Stratified sampling may be suitable. 17 packets of beef, 12 packets of chicken, 15 packets of beef with rice, 6 packets of chicken with rice would be randomly selected from all the packets produced during each shift.

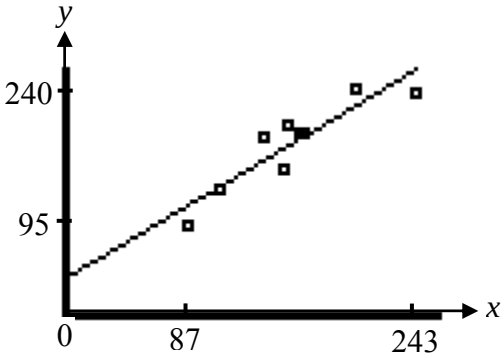
Qn	Solution
7	Probability
(i)	 P: packed properly P': not packed properly $\frac{7}{20} \times 0.01 + \frac{1}{5} \times 0.03 + \frac{3}{20} \times 0.03 + \frac{3}{10} \times \frac{p}{100} = 0.02 \quad p = 2$

(ii)	Required probability = $\frac{7}{20} \times \frac{99}{100}$ = 0.3465
(iii)	Required probability = $\frac{0.3465}{1 - 0.02}$ = 0.354
	Required probability = $(0.98)^{10}$ = 0.817

Qn	Solution
8	Hypothesis Testing
(i)	$\bar{x} = 2 + \frac{408}{80} = 7.1$ $s^2 = \frac{1}{79} \left(\sum (x-2)^2 - \frac{\sum (x-2)^2}{80} \right)$ $= \frac{1}{79} \left(2116 - \frac{408^2}{80} \right) = 0.4455696 = 0.446 \text{ (3.sf)}$
(ii)	Since $n = 150$ is large, by Central Limit Theorem, $\bar{X} \sim N \left(7.1, \frac{0.668^2}{\sqrt{80}} \right) \text{ approximately.}$ $H_0: \mu = 7.34$ $H_1: \mu < 7.34$ Test Statistic: $Z = \frac{\bar{X} - \mu}{S / \sqrt{n}}$ Level of significance: 5% Critical Region: Reject H_0 if $p\text{-value} < 0.01$ Assuming H_0 is true, from the GC, $p\text{-value} = 0.0006503 = 0.000650 \text{ (3s.f.)}$. Since $p\text{-value} = 0.000650 < 0.01$, we reject H_0 and conclude that there is significant evidence, at 1% level, that the long jumper had over-estimated his ability. Not necessary. Since sample size of 80 is large and Central Limit theorem can be applied.

Qn	Solution
9	Binomial & Approx
	Let X be the no of defective articles out of 50. $X \sim B(50, 0.02)$
(i)	$E(X) = 50(0.02) = 1$
(ii)	$P(X \geq 2)$ $= 1 - P(X \leq 1)$ $= 0.26423$ $= 0.264(3sf)$
(iii)	$P(X \leq 4 X \geq 2)$ $= \frac{P(2 \leq X \leq 4)}{P(X \geq 2)}$ $= \frac{0.26102}{0.26423} = 0.988(3sf)$
	Let Y be the no of samples out of 100 that have at least 2 defective articles. $Y \sim B(100, 0.26423)$ $n = 100$ is large, $np = 26.423 > 5$, $n(1-p) = 73.577 > 5$ $Y \sim N(26.423, 19.441)$ approx $P(Y \geq 20) = P(Y > 19.5)$ using continuity correction $= 0.942(3sf)$

Qn	Solution
10	Correlation and Regression
(i)	Using the GC, $r = 0.155$ (3 s.f.) As r is approximately zero, x and y may not be linearly correlated.
(ii)	

(iii)	As (236, 27) is an outlier, the comment in (i) should be amended. If (236, 27) is removed, the new $r = 0.903$ which indicates a strong positive linear correlation between x and y. OR If (236, 27) is removed, the points lies close to the regression line of y on x, which seems to indicate that x and y are linearly correlated. 
(iv)(a)	Using the GC, $y = 39.945 + 0.89779x = 39.9 + 0.898x$
(iv)(b)	Using the GC, $x = -7.1970 + 0.90665y = -7.20 + 0.907y$
(v)(a)	Using $x = -7.1970 + 0.90665y$ When $y = 183$, $x = -7.1970 + 0.90665(183) = 159 \quad (3 \text{ s.f.})$
(v)(b)	Using $y = 39.945 + 0.89779x$ When $x = 257$, $y = 39.945 + 0.89779(257) = 271 \quad (3 \text{ s.f.})$
(vi)	For (v)(a), $y = 183$ is within the data range of y. Hence the estimated test score is reliable. For (v)(b), $x = 257$ is outside the data range of x. Hence the estimated test score may not be reliable.

Qn	Solution
11	Normal Distribution with Central Limit Theorem
	Let F = the mass, in kg, of a full packet of flour. $F \sim N(25, 1.5^2)$
(i)	$P(F > 27) = 0.091211 = 0.0912 \quad (3 \text{ s.f.})$ $P(\text{lightest of the 3 packets is more than 27 kg})$ $= P(\text{all 3 packets are each more than 27 kg})$ $= P(F > 27)^3$ $= 0.091211^3$ $= 0.000759 \quad (3 \text{ s.f.})$
(ii)	$F_1 + F_2 + F_3 + F_4 + F_5 \sim N(125, 11.25)$

	$P(F_1 + F_2 + F_3 + F_4 + F_5 < 130) = 0.932 \text{ (3 s.f.)}$
(iii)	$\bar{F} = \frac{F_1 + F_2 + \dots + F_n}{n} \sim N\left(25, \frac{1.5^2}{n}\right)$ $P(\bar{F} > 26) < 0.1$ $P\left(Z > \frac{\sqrt{n}}{1.5}\right) < 0.1$ $P\left(Z \leq \frac{\sqrt{n}}{1.5}\right) > 0.9 \Rightarrow \frac{\sqrt{n}}{1.5} > 1.28155$ $\therefore n > 3.6953$ Least n is 4
(iv)	$\bar{F} = \frac{F_1 + F_2 + \dots + F_{25}}{25} \sim N(25, 0.3^2)$ Let D = the weight, in kg, of a damaged packet of flour. $E(D) = 13$, $\text{Var}(D) = 2.5^2$ Since 55 is large, by Central Limit Theorem, $\bar{D} = \frac{D_1 + D_2 + \dots + D_{55}}{55} \sim N\left(13, \frac{2.5^2}{55}\right) \text{ approximately.}$ $P(\bar{F} - \bar{D} > 11) \quad \bar{F} - \bar{D} \sim N(12, 0.20364) \text{ app}$ $= 0.987 \text{ (3 s.f.)}$